

✱ Calculus 1 ✱

(Intervals and inequalities) sw

Definition sw

A closed interval, denoted by $[a; b]$ consist of all real numbers x for which $a \leq x \leq b$.

An open interval, denoted by (a, b) consist of all real numbers x for which $a < x < b$.

Example (1) :-

1] Write inequality using interval notation :-

1- $1 \leq x \leq 3$

$[1, 3]$

~~2- $x > 5$~~

2- $-4 < x < 0$

$(-4, 0)$

3- $1 \leq x < 4$

$[1, 4)$

(1)

2] Write the following using inequality notation :-

1- $[1, 4]$
 $1 \leq x \leq 4$

2- $(-\infty, -3]$
 $x \leq -3$

(Inequalities) :-

An equation says that two expressions are equal, while inequality says that one expression is greater than, greater than or equal to, less than or less than or equal to, another, as with equation, a value of the variable for which the inequality is true is a solution of the inequality, the set of all such solutions makes up the solution set of the inequality.

(2)

(Two inequalities with same solution set are equivalent) :-

Properties of inequality :-

1\ Addition for a real number a, b and c :-

(i) If $a < b$ then $a + c < b + c$

(ii) If $a > b$ then $a + c > b + c$

2\ Multiplying inequality by positive number :-

(i) $a < b$ then $ac < bc$

Example :- $3 < 7$

Multiply both side by 2 $\Rightarrow 6 < 14$

3\ Multiplying inequality by negative number the effect of multiplying

both side by negative numbers the direction of the inequality symbol is reversed $9 > 2$

Multiplying by $-4 \Rightarrow -36 < -8$

(3)

Note 8-

Similar Properties are valid if $<$ replaced with $>$, $\leq$ or $\geq$.

Example 8-

Solve the following inequality 8-

1) $2x < 6$

Sol 8-

$$\frac{1}{2}(2x) < \frac{6}{2}$$

$$\Rightarrow x < 3, (-\infty, 3)$$

2) $x > 12$
-3

Sol 8-

$$\frac{(-3)x}{-3} < \frac{(-3)12}{-3}$$

$$\Rightarrow x > -36, (-36, \infty)$$

(Absolute Value and Distance) 8-

We start with a geometric definition of absolute value. If a is the coordinate of a point on a real number line, then the distance from the origin to a is represented by $|a|$ and is referred to as the absolute value of a .

Definition 8-

$$|x| = \begin{cases} x & \text{if } x \geq 0, & |4| = 4 \\ -x & \text{if } x < 0, & |-3| = 3 \end{cases}$$

Note 8-

The absolute value of a number is never negative.

Properties of Equations and Inequalities Involving $|ax+b|$

for $P > 0$:

1- $|ax+b|=P$ is equivalent to
 $ax+b=P$ or $ax+b=-P$

2- $|ax+b| < P$ is equivalent to
 $-P < ax+b < P$

3- $|ax+b| > P$ is equivalent to
 $ax+b < -P$ or $ax+b > P$

Example :- Solve and write Solutions in both inequality and interval notations

1] $|3x+5| = 4$

Sol:-

$$3x+5 = \pm 4 \Rightarrow 3x = -5 \pm 4$$

$$\Rightarrow x = \frac{-5 \pm 4}{3} \Rightarrow x = -3, -\frac{1}{3}$$

or $\{-3, -\frac{1}{3}\}$

(6)

$$2) |2x-1| < 3$$

Sol:-

$$|2x-1| < 3 \Rightarrow -3 < 2x-1 < 3$$

$$\Rightarrow -2 < 2x < 4 \Rightarrow -1 < x < 2$$

or $(-1, 2)$

$$3) |2x-1| \geq 3$$

Sol:-

$$2x-1 \leq -3$$

$$\text{or } 2x-1 \geq 3$$

$$2x \leq -2$$

$$\text{or } 2x \geq 4$$

$$x \leq -1$$

$$\text{or } x \geq 2$$

$$(-\infty, -1] \cup [2, \infty)$$

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